We Will Find You: An Edge-Based Multi-UAV Multi-Recipient Identification Method in Smart Delivery Services

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Abstract. Unmanned aerial vehicle (UAV) is increasingly becoming a promising solution for last-mile delivery in smart logistics, and multi-UAV scenarios have become increasingly common. In multi-UAV delivery services, the ability to accurately and efficiently identifying multiple target recipients is a critical issue. Face recognition has been widely used in UAV delivery services but since the UAVs need to passively wait for the target recipient to arrive at the designated location, the efficiency of identification is undesirable. Furthermore, as one UAV is intended only for one recipient, there is no collaboration between multiple UAVs at the same location which could otherwise reduce the computational and communication time for some common tasks. To address the aforementioned issues, in this paper, we propose an active multi-UAV multi-recipient identification method (named MM4ID) in edge-based smart delivery services, based on person re-identification (ReID) technology. Specifically, multiple UAVs scan the recipient's activity area and transmit their video streams to a nearby edge server. In the meantime, a multiple object tracking (MOT) algorithm is performed at the edge server to track the recipients and obtain one image for each recipient from the video streams to reduce duplicated data. We perform person re-identification algorithm on the obtained recipient images to determine the location of the recipient, and then we use the face recognition algorithm to confirm the identity of the recipient. Finally, successful identification results including the target recipients and their locations are sent to all UAVs via broadcast so that each UAV can complete their final delivery task. Experimental results based on a real-world edge-based smart UAV delivery service system successfully demonstrate the effectiveness of our proposed method and yield better performance compared with other representative solutions.

Keywords: Face Recognition · UAV Delivery · Person Re-identification · Multiple Object Tracking

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1 Introduction

In recent years, smart logistics has become an effective means to improve delivery efficiency and customer satisfaction [1,2]. UAVs delivery service has become a promising solution for last-mile delivery in smart logistics, and scenarios of multiple UAVs for logistics delivery are becoming increasingly common [3–6]. In multiple UAVs delivery systems, fast identification of target recipients with high accuracy is a key challenge for satisfactory user experience [7]. A more effective and cost-efficient solution is to use face recognition to confirm the identity of the recipient [8].

At present, the target recipient identification method based on face recognition algorithm still has some challenges, particularly in the UAV logistics scenario [9,10]. On the one hand, although recognition can be achieved with only face recognition method, its efficiency is low because it requires the recipient to be at a designated location. Until then the UAV needs to passively hover at the designated location and wait for the target recipient to arrive at the designated location, which will incur time costs, energy consumption and reduce the quality of service (QoS). On the other hand, the recipient should be able to receive the delivery from the UAV as soon as it arrives at the designated location. The identification should be done actively by the UAV before the recipient arrives at the designated location. Although there are some existing research works, they all identify the recipient after the recipient has arrived at a designated location, which can be very inefficient. Face recognition is performed after the recipient location is determined, which undoubtedly increased the practicality of UAVs to identify the recipient [11,12]. Pose recognition algorithm is used to proactively find the target recipients' location, but the algorithm can be time-consuming [11]. Person re-identification algorithm is used to proactively find the target recipients' location [12], but it does not consider the highly redundant data generated by the terminal device. In the frame-by-frame data processing, there are several duplicated data between frames, and the data of one pedestrian is processed several times.

To address the aforementioned issues, in this paper, we propose an active multi-UAV multi-recipient identification method (named MM4ID) in edge-based smart delivery services, based on person re-identification. First, the UAVs capture video streams in the recipient's activity area and transmit the data to a nearby edge server. Second, for the video streams captured by UAVs, the multiple object tracking (MOT) [13] algorithm named YOLOv5-DeepSORT is used to track the person and obtain one image for each recipient to reduce duplicated data. Third, we perform person ReID algorithm [14] on the obtained recipient image to determine the location of the recipient. Fourth, we use the face recognition algorithm to confirm the recipient. Finally, successful identification results are sent to all UAVs via broadcast and the corresponding UAV completes the logistics delivery. Experimental results in a simulation of a real smart UAV delivery system demonstrate the effectiveness of the method.

The rest of this paper is organized as follows. Section 2 presents some related work. Section 3 provides an overview of the framework and describes the detailed models and algorithms used in our proposed framework. Section 4 presents the experimental results of the real-world UAV delivery system. Finally, Sect. 5 concludes the paper and indicates future work.

2 Related Work

In the process of UAV delivery with edge computing, the ability to efficiently identify the recipient is a popular research topic [15–17]. There are already some methods based on face recognition algorithms to solve the target recipient identification problem in the UAV logistics delivery scenario. Shao et al. used a face detection algorithm to detect the recipient's faces and then input the detected faces into the face recognition network for identity confirmation [18]. However, since recipients often keep moving and their positions are mostly random in the real world, this approach cannot ensure satisfactory face recognition in UAV delivery scenarios. Shen et al. proposed a face scoring mechanism and a UAV flight control algorithm that enables the UAV to adjust its orientation according to the recipient's pose [19], thus enabling automatic capture of the recipient's frontal for face recognition. Since this method ensures that the UAV always captures the frontal aspects of the recipient, it can improve its effectiveness in UAV delivery scenarios. However, the UAV requires a lot of time and energy to adjust its flight direction to capture the frontal aspect of the recipient. Therefore, this method is not energy efficient for UAVs, which makes it impractical to use in realistic scenarios. Although the recognition accuracy can be achieved in the aforementioned methods, the efficiency of these methods is low because face recognition algorithm requires the recipient to be in a designated location. The UAV needs to passively wait for the target recipient to arrive at a designated location, which will bring time as well as energy consumption and reduce the quality of service (QoS). To solve this problem, Gao et al. proposed a method to first filter some recipients' image by processing the video frame by frame through pose recognition to proactively find the target recipients' location, and then complete the identity confirmation by capturing the frontal face [11]. This method improves the applicability of face recognition, but the time complexity of the pose recognition algorithm is usually high. Therefore, the efficiency of the whole process cannot be guaranteed and powerful computing hardware is required. Zhang et al. performed target recipient identification by person ReID algorithm to complete logistics delivery [12], but the accuracy of single-person ReID is low in practical scenarios, which is difficult to meet the high precision requirements of logistics delivery. In addition, it does not consider the highly redundant data generated by the terminal device.

Therefore, there is a need to design a method for the UAV to actively and efficiently find the target recipient thereby improving the efficiency of target recipient identification.

3 Framework and Algorithms

3.1 Framework

The UAV logistics delivery system in edge computing consists of three main components: the cloud server, the edge server, and the UAV. The cloud server stores all users' information (including their registered face images and full body images) and continuously provides intelligent services to the whole system. The edge server has sufficient computing resources, and it is closer to the end device than the cloud server, which can effectively reduce the data transmission time. AI models and algorithms (such as MOT, person ReID and face recognition algorithms) can be deployed on the edge server. UAVs are used as end devices for transportation systems, such as sensors, cameras, computing devices, batteries, and other resources on board, mainly for flight and cargo-related tasks.

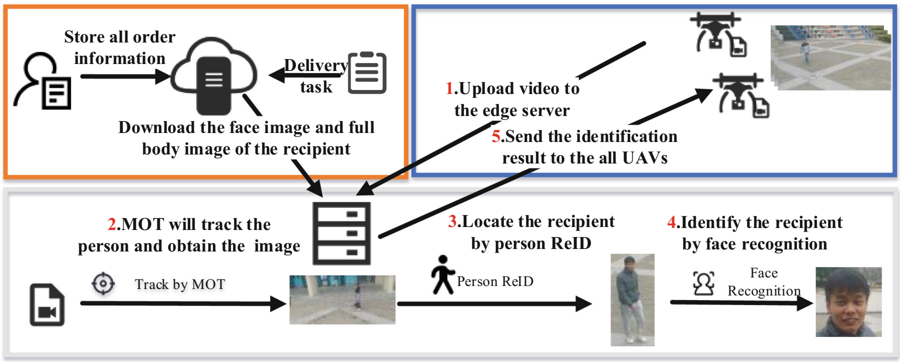


Fig. 1. Processing flow of MM4ID.

The processing flow of MM4ID is shown in Fig. 1. When there is a delivery task, the edge server will request the data of the target recipient on the cloud server and arrange the UAV to deliver the goods. After the UAV arrives in the recipient's activity area, the video captured by the UAV is analyzed frame by frame to determine the location and identify the recipient. The recipient identification process is divided into four main steps as follows.

Step 1: The UAV captures the original video frames in the recipient's activity area and sends the video data to the edge server.
Step 2: The video data captured by the UAV will then be processed on the edge server. The edge server first assigns an identity document (ID) to each pedestrian in the video and tracks and locates it by MOT. For each ID's pedestrian, obtain one image to reduce duplicate data.
Step 3: The recipient is identified using the person ReID with the recipient's picture to determine the recipient's location. After determining the existence

of a pedestrian matching the picture of the recipient by the person ReID, the UAV approaches this pedestrian and shoots the front face images of the pedestrians.
Step 4: The face recognition algorithm is used to confirm whether the face of the pedestrian determined by the person ReID matches the image of the recipient's face downloaded from the cloud server.
Step 5: The edge server broadcasts the identification result to all UAVs after successful matching. After the corresponding UAV receives the identification result, the UAV will land near the target recipient and complete the delivery.

3.2 Algorithms

Multiple Object Tracking (MOT). The MOT used in this paper is YOLOv5-deepsort [20]. YOLOv5 is used as the recipient detector, and deepsort is used to track the detected target recipient. As an object detection algorithm, YOLOv5 can efficiently detect pedestrians and YOLOv5 network structure is shown in Fig. 2 [21], which consists of four components: Input, Backbone, Neck, and Prediction.

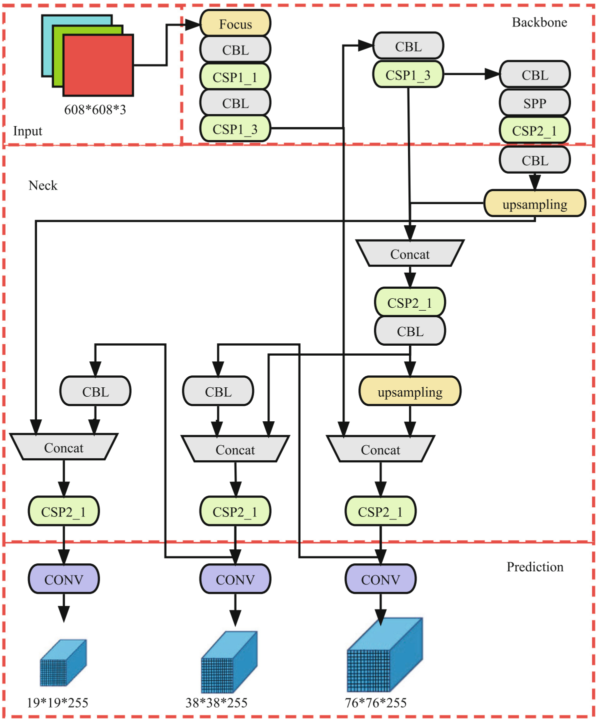


Fig. 2. YOLOv5 network structure.

The Input component completes the preprocessing of the image, including Mosaic data enhancement, adaptive anchor frame calculation, and adaptive image filling. The Backbone component adopts a focused structure. It is responsible for the feature extraction of the input image. The Neck component is the feature pyramid network (FPN) and path aggregation network (PAN) structure. It uses a CSP2 structure borrowed from the cross stage partial network (CSP-net) design, which enhances the ability of network feature fusion. The Prediction adopts the GIoU loss function to achieve faster and better convergence. The loss function is defined as follows:

$$Loss_{GIoU} = 1 - \frac{A^p \bigcap A^g}{A^p \bigcup A^g} + \frac{C - A^p \bigcup A^g}{C} \tag{1}$$

Where A^p denotes the prediction box, A^g denotes the ground truth box, and C denotes the smallest outer rectangle of the two boxes. The main problem involved in detection-based tracking is the problem of prediction, matching, and updating after detection. As a tracking algorithm, deepsort also has good target tracking performance in the presence of occlusion and the process is shown in Fig. 3 [22].

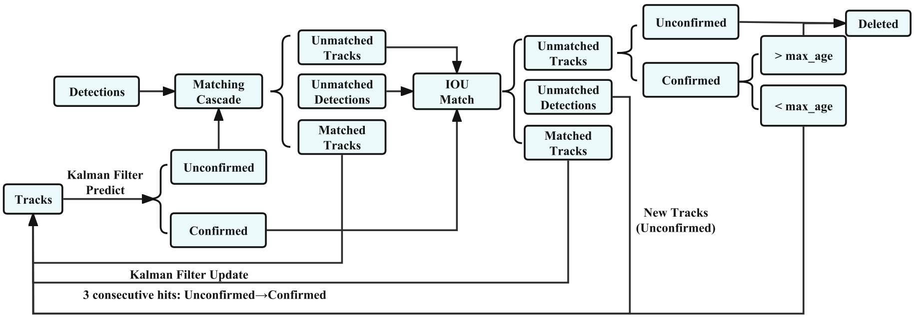


Fig. 3. Processing flow of deepsort.

The target is predicted by Kalman filtering, and the Mahalanobis distance between the obtained predicted target and the current target is calculated. When its value is less than the specified threshold, it means that the predicted target is correct. The Mahalanobis distance is calculated as follows:

$$d^{(1)}(i,j) = (d_j - y_i)^T S_i^{-1} (d_j - y_i) \tag{2}$$

$d^{(1)}(i,j)$ denotes the motion match between the i-th detection target and the j-th trajectory. S_i is the covariance matrix of the observation space of the current moment predicted by the trajectory after using the Kalman filter, y_i denotes the predicted observations of the trajectory at the current moment, and d_j is the position information of the j-th detection target. To prevent the ID from changing constantly, the minimum cosine value between the predicted target and the feature vector of the predicted target contained in the trajectory is used as the apparent matching degree to measure the prediction accuracy. The expression is calculated as follows:

$$d^{(2)}(i,j) = min\left\{1 - r_j^T r_k^{(i)} \mid r_k^{(i)} \in R_i\right\} \tag{3}$$

Where r_j denotes one feature vector corresponding to each detection block d_j, $|r_j| = 1$, and r_k is the feature vector of the most recent successful association stored for each tracking target. Finally, the final measure is as follows:

$$c(i,j) = \lambda d^{(1)}(i,j) + (1-\lambda)\, d^{(2)}(i,j) \tag{4}$$

Where λ is the weight parameter and finally the Hungarian algorithm is introduced to detect whether a target in the current frame, is the same as a target in the previous frame. To avoid a track being obscured for a period of time or after losing frames for a long time, cascade matching is used to ensure that the predicted target is the target before being obscured or before losing frames for a long time.

Person Re-identification (ReID). In this paper, person ReID is used as a method for recipient localization. The backbone network of person ReID is ResNet50 and the training process is shown in Fig. 4 [14].

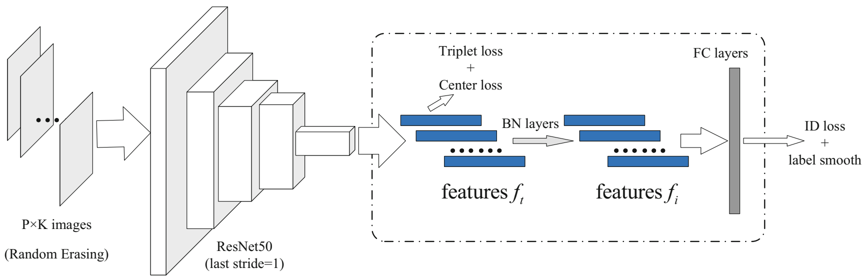


Fig. 4. Training process of person ReID.

Random Erasing Augmentation (REA) is used in the training process to reduce the degree of model overfitting, so it can improve the performance of the model, and the last stride down sampling operation in the backbone is removed to increase the size of the feature map. A batch normalization (BN) layer is added after the features f_t. The features before BN layer are f_t and after BN is f_i, which makes the ID loss easier to converge.

Face Recognition. The face recognition model is trained by the deep convolutional network [8], and triplet loss is used as the loss function. The loss function is as follows:

$$\varphi(A, P, N) = \max\left(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0\right) \tag{5}$$

Where $f(A)$ denotes the face recognition model we need to train. The 128 feature vectors of image A can be calculated through $f(A)$. A is an image of a person, P is another image of the person, N is an image of another person, and α is the spacing distance. The specific training process is to divide the training picture into triples, make A and P as close as possible, and A and N as far as possible. Through the trained face recognition model, the feature values F_i can be extracted, which is composed of 128 feature vectors x_{in} $(n = 1,\ 2,\ \ldots,\ 128)$ in the face image. For face matching, the ED (Euclidean distance) is used as the similarity measure between the face feature values. The ED is calculated as follows:

$$ED(F_i, F_j) = \sum_{n=1}^{128} \sqrt[2]{(x_{in} - x_{jn})^2} \tag{6}$$

F_i and F_j represent the face feature values of two images. If the calculated $ED(F_i, F_j)$ is less than the threshold value we set, it is determined that the same person is identified, otherwise, it is not.

4 Evaluation

In this section, we present a case study and experimental results to verify the effectiveness of our proposed method. The videos used in the experiments were collected by using the UAV (DJI Matrice 600 Pro) and all experiments were run on the computer with an 11th generation Intel (R) Core (TM) i5-11400 processor and 16 GB of RAM, with a 3060Ti 8G GPU and Windows 10 (64-bit) operating system. The data and algorithms used in our experiments are made available online on GitHub: https://github.com/GRY-Code/MM4ID.

4.1 Case Study

We introduce MM4ID with a real-world case study. We suppose that there are two delivery tasks T_1 and T_2. UAV_1 and UAV_2 denote the UAVs corresponding to T_1 and T_2. U_1 and U_2 denote the target recipient corresponding to T_1 and T_2, $Image\_Face_1$ and $Image\_Face_2$ denote the face images of U_1 and U_2, $Image\_Body_1$ and $Image\_Body_2$ denote the body images of U_1 and U_2, ES denotes the edge server, and CS denotes the cloud server. j denotes the ID of the pedestrian we assign, and P_j denotes the pedestrian whose ID is j. ES in the area will first launch a data request to the CS to download the $Image\_Face_1$, $Image\_Face_2$, $Image\_Body_1$ and $Image\_Body_2$. The corresponding processing flow is as follows.

Firstly, the ES arranges UAV_1 and UAV_2 to fly to the recipient's activity area to deliver the goods. The UAV_1 and UAV_2 send the captured video streams to the ES as shown in Fig. 5. Secondly the ES first uses the MOT to track and locate the pedestrian in the video and assign ID as shown in Fig. 6.

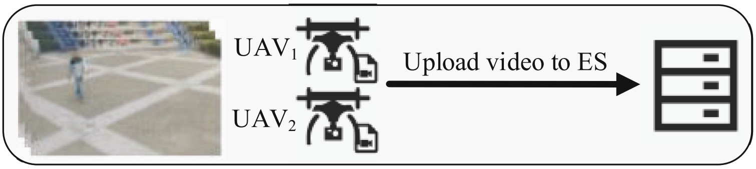


Fig. 5. UAV capture data.

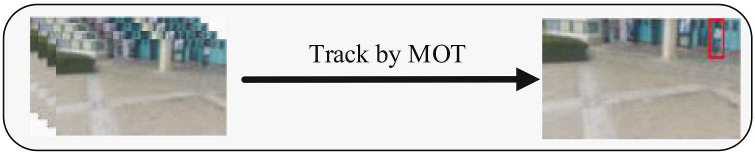


Fig. 6. Track by MOT.

Thirdly each numbered pedestrian will be matched once with $Image\_Face_1$ and $Image\_Face_2$ by person ReID. If P_m captured by UAV_1 is matched with $Image\_Body_2$ as shown in Fig. 7. UAV_1 will move to the vicinity of P_m and capture the positive frontal face.

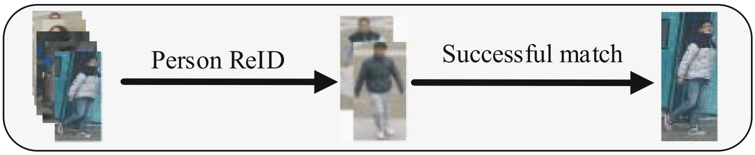


Fig. 7. Locate by person ReID.

Fourthly, U_2 is matched by face recognition as shown in Fig. 8. If the face picture of P_m taken by UAV_1 matches with $Image\_Face_2$ through face recognition, the P_m is determined to be the U_2.

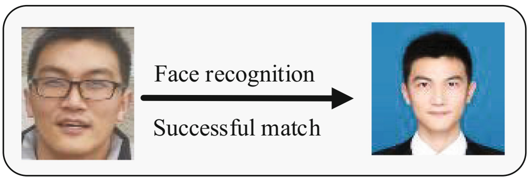


Fig. 8. Face recognition to determine identity.

Finally, the ES will broadcast the successful identification result is sent to UAV_1 and UAV_2 as shown in Fig. 9. When UAV_2 receives the broadcast message, it will fly to the location corresponding to P_m and lands to complete the logistics delivery. UAV_1 continues to use the above steps to find U_1.

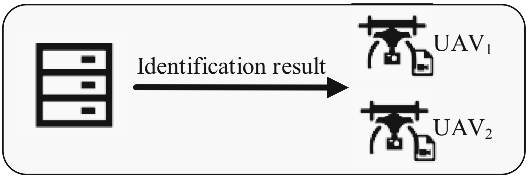


Fig. 9. ES broadcasts the result to UAV_1 and UAV_2

If the target recipient is not found when the recipient's activity area search has been completed, it is determined that the recipient is not within the receiving area. The corresponding UAV will then fly back to the logistics station. When the recipient initiates the next delivery request, another delivery will be made.

4.2 Experiments

We perform three sets of comparison experiments. The first set of experiments evaluates the efficiency of the task performed in cloud server and edge server. In the second set of experiments, we compare the efficiency of our method with two other methods, a baseline method using only face recognition [18] and Edge4Sys which combines pose recognition and face recognition [11]. The third set of experiments is the ablation experiments.

In the first set of experiments, we compare the time for recognition tasks processed in the cloud server and in the edge server as the number of target recipients increases when processing seven video frames per second using MM4ID.

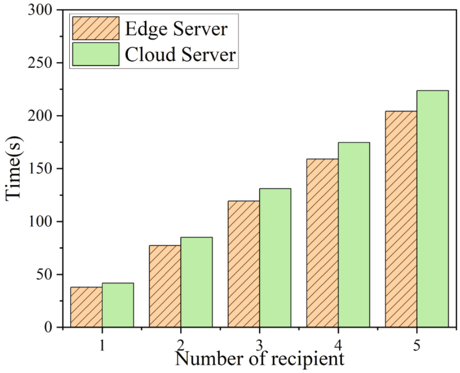
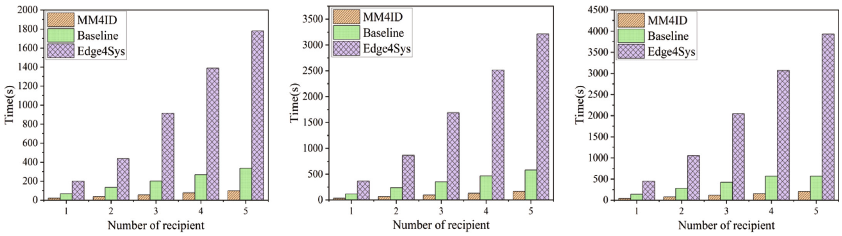


Fig. 10. Comparison between edge server and cloud server.

As shown in Fig. 10, the time for processing tasks in the edge server is less than the time of processing tasks in the cloud server. This is because when data is processed in the cloud server, the data captured by the UAV needs to be transferred to the cloud server, and the time of the transfer affects the overall task processing rate. The edge server is located near the UAV so the time for data transfer is low, making it more suitable for UAV logistics delivery scenarios.

In the second set of experiments, we compare the efficiency of these three methods. The baseline method is a passive target recipient recognition approach using only face recognition algorithm. Edge4Sys is using tiny-YOLO [23] to capture pedestrian images and pose recognition to locate recipients before face recognition algorithm. We compare the time consumption of the three methods for processing the number of video frames per second of 3, 5, and 7 for the number of target recipients of 1, 2, 3, 4, and 5.

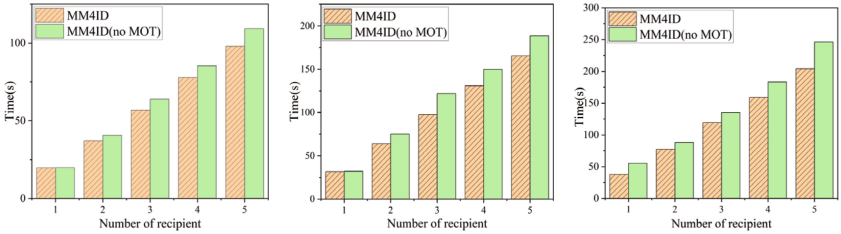


3 frame/s 5 frame/s 7 frame/s

Fig. 11. Comparison of the efficiency of different methods.

The efficiency comparison results of the three methods are shown in Fig. 11. Edge4Sys has the lowest efficiency because pose recognition takes more time to process the data, and there is also the case where the data of the same person is processed multiple times. These factors significantly affect task processing efficiency. The baseline method takes less time because it only uses face recognition and does not use some other methods of recipient localization such as pose recognition. However, to only use face recognition in a practical delivery scenario, the recipient needs to stand at a designated location and wait, the UAV passively comes to find the recipient. If the location is not designated, or if the recipient is moving, and face recognition has certain requirements for the pixel density of the photographed face, the UAV will need to fly over a wide area to perform face matching, making it impractical. The efficiency of our proposed method is optimal because the data requiring person ReID is effectively reduced subsequently through MOT. With the person ReID, our proposed method can accurately locate the recipient, which is highly efficient in UAV logistics delivery. The final face recognition also further ensures the accuracy of identifying the recipient.

The accuracy of existing face recognition algorithm reached 99.68% [8], and their extremely low error rate made them widely used for mobile phone unlocking, access control, etc. The methods we compare all use face recognition as the final means of identification. The accuracy of different methods is the same. Therefore, in this experiment, we only consider the case of successful delivery, and for the scenario of extremely low probability of logistics delivery errors, we will consider it in our future work.



3 frame/s 5 frame/s 7 frame/s

Fig. 12. Comparison of the efficiency of with and without MOT.

In the third set of experiments, we compared the time consumption of MM4ID and MM4ID without MOT to process 3, 5, and 7 video frames per second with the number of target recipients of 1, 2, 3, 4, and 5. As shown in Fig. 12, using MOT is more efficient than without using MOT. This is because the use of MOT can effectively remove duplicated data and improve recognition efficiency, and the effect will be more obvious when the number of recipients increases or the number of images to be processed per second increases.

5 Conclusions and Future Work

Face recognition has been widely used for person identification in many intelligent Internet of Things systems, including intelligent UAV delivery systems for smart logistics. However, the efficiency of face recognition methods in edge-based smart UAV delivery services are still open issues. In this paper, we proposed an active multi-UAV multi-recipient identification method named MM4ID. Specifically, our proposed method can effectively coordinate the computation and communication between multiple UAVs and the edge server, and take full advantage of the multiple object tracking algorithm, person re-identification model, and the face recognition method to improve the overall efficiency of the recipient identification process. Experimental results based on real images captured by UAVs demonstrated that our proposed method can achieve the best results in recognition efficiency compared with other representative solutions.

There are still some future research directions in this area. For example, in this paper, we only considered the case in which the recipient is correctly identified, but we have not discussed the rare but complex cases where some errors may occur. This could potentially be handled as exceptions with human intervention. In addition, we will consider integrating data security measures for personal ReID (such as biometric information protection) to further improve the identification process.

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